AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM352
Web, Mobile and Cloud Technologies

Midterm Examination (MTA)
Spring Semester 2025/2026

Date: 15/April/2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2 Hrs**

Instructions:

- Total Marks: 45 {will be adjusted to 30 marks}
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Basic Calculator** is allowed.

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PART I (5 Marks)

Answer all the questions

Choose the right answer

1) Which of the following protocols related to network transport layer of the web service architecture
   - A) TCP, UDP
   - B) HTTP, SMTP
   - C) IP, ICMP
   - D) PPP, FTP
2) The objects are organized in the HTML DOM as .....
   - A) Queue
   - B) Class-wise
   - C) Hierarchy
   - D) Stack
1) Which method is used to display a message in a dialog box using javascript?
   - A) alert()
   - B) prompt()
   - C) message()
   - D) console.log
4) In a RESTful web service the request is transported as
   - A) XML
   - B) HTML
   - C) JSON
   - D) URI
5) HTTP method of submitting data to server
   - A) post
   - B) submit
   - C) upload
   - D) put

PART II (20 Marks)

Answer all the questions

1) Compare between Open Systems Interconnection (OSI) model and TCP/IP Model in terms of the used layers [4 Marks]
2) Describe the web service components and operation. [6 marks]
3) Compare symmetric and asymmetric key encryption methods, highlighting one advantage and one disadvantage of each. [6 marks]
4) Define the three components of the CIA triad and match each one to an appropriate class of security threat. [4 Marks]

PART III (20 Marks)

Answer all the questions

1) When consuming the calculate method of the currency service by typing in the address part of the web browser [6 marks]

   a. Complete the address to be typed
   /webresources/...../.....? from=**&amount=**

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b. Complete the following RESTful web service, by writing the service annotations

```
import com.Conversions;
    @Path("Currency")
    public class Currency {
       @-----
       @Path("calculate")
       @Produces(MediaType.TEXT_HTML)
      public String Calculate(@wcb Param("------")String x,@-------
      Param("---------")String y) {
         return Conversions.convert(x,y);}
      }
```

2) Provide JSON representation for data in the following table: [6 marks]

Time	Flight	From	AirLine	Aircraft
3:25 PM	TK92	Istanbul (IST)	Turkish Airlines	A321 (TC-JSZ)
3:40 PM	GF183	Bahrain (BAH)	Gulf Air	A321 (A9C-CE)
3:45 PM	SV1035	Riyadh (RUH)	Saudia	A333 (HZ-AQ30)

3) Write an HTML page with a form containing two textfields for student name and his/her gender. On submission, the values are sent to the URI "/collecting" [4 marks]

4) Using the Caesar Cipher encryption technique with a key value of 8, encrypt the following plaintext: "Give"

Show the step-by-step shifting process for each character and write the final ciphertext. [4 marks]

End of questions



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